

ALTR: what changed, and why it had to

From accounting profits to asset-level free cash flow — the model updates, the evidence behind each one, and what comes next.

Facts as of model version 7b7546a · defaults cited from runtime config (conf/base)

Note: ALTR (Asset-Level Transition Risk) is our next-generation model, an intentional rebrand from TRISK to own the IP narrative. Everything quantitative in this deck is version-stamped — no figures from memory.

TRISK values earnings. Markets value cash.

TRISK's net-profit block is revenue minus a per-MWh cost, discounted. It is an earnings construct — not the cash a firm can distribute.

NO INVESTMENT

No CapEx, no replacement, no decommissioning — the cycles that drive power-sector cash flows.

FRAGILE MARGIN DATA

Net profit margins from Refinitiv Eikon: gap-filled averages, unreliable at asset level.

EVERYTHING SCALES WITH OUTPUT

All costs production-proportional. Fixed O&M on a stranded plant? Invisible.

Say: valuation practice (FCFF/FCFE) is explicit — use free cash flows, not accounting profits or EBITDA alone. Using earnings double-counts or ignores parts of the cash-flow stream. This is the motivation for every change that follows.

Asset-level free cash flow, bottom-up

Each firm is the sum of its physical plants — location, capacity, age, emission factor. AR6 trajectories hit each asset; cash flows aggregate up.

$$\text{Revenue} = Q \times \text{price} - \text{Fuel} = Q \times \text{fuel}/\eta - \text{Fixed O\&M} = K \times \text{fom} - \text{Carbon (net of passthrough)} = \text{EBITDA} -$$

$$\text{CapEx streams} = \text{FCFF} \rightarrow \text{DCF} \rightarrow \text{NPV}$$

Output: valuation shock = ΔNPV between below-2°C shock and current-policies baseline. Q = production volume (MWh), K = capacity (MW).

Say: no corporate tax, no depreciation, no working capital — deliberately excluded; we don't have that accounting data at asset level and say so. The published paper's claim that asset-level data lowers estimated transition cost by ~7% is kept as the paper's claim, not generalized.

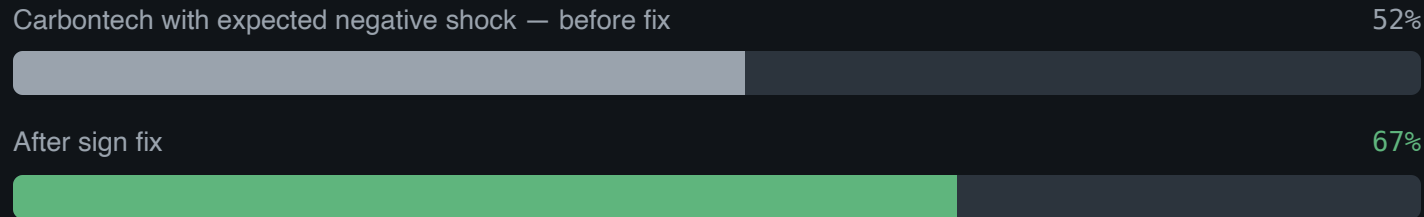
Three CapEx streams — two on, one deliberately off

STREAM	RULE	DEFAULT	WHY
Replacement	2% of installed MW / year	ON	EPRI / Lazard utility maintenance benchmark; retiring assets excluded
Decommissioning	50% of CapEx/MW at retirement	ON	Demolition, remediation, site restoration
New-build	$\Delta K^+ \times \text{CapEx/MW}$	OFF	IAM O&M (REMIND, WITCH, POLES) already bundles annualized capital — including it double-counts

Say: new-build OFF is a choice, not an omission. Turning it on with AR6 O&M data charges green builders twice and skews results against exactly the assets that should win. A provider-level audit (future work) decides where it can be safely enabled.

Decommissioning is a cost, not a payday

Early versions treated retirement as positive scrap income. Real plant retirement is a net cash outflow. Fixing the sign was the single largest direction lever we found.



Say: a stranded coal plant has no salvage market; what it has is demolition and site-remediation liabilities (ARO-style). Charging 50% of CapEx/MW as an outflow is crude but directionally right — technology-specific ratios are queued as future work.

Four upgrades beyond the discount rate

1 · TECHNOLOGY RISK SPREADS

Carbontech **+100bps**, green **-50bps**, both scenarios at 7%. Bolton & Kacperczyk carbon premium; Shell 2024 discloses a 150bps O&G-vs-renewables spread.

2 · STRANDING-AWARE TERMINAL VALUE

3 loss-making final years $\rightarrow TV = 0$. Profitable-but-declining fossil $\rightarrow$ 10-year annuity, not perpetuity. Green $\rightarrow$ standard perpetuity. (Gourdel 2024 abandonment logic.)

3 · TERMINAL CASH FLOW NORMALIZATION

Average of last 3 years, not the final year — one CapEx spike can no longer erase a terminal value. (Damodaran / McKinsey practice.)

4 · SPLIT TERMINAL GROWTH

Fossil 0%, green 2%. At $r=7\%$, flat 2% growth means a $20.4\times$ terminal multiple vs $14.6\times$ at 0% — a ~28% overvaluation of declining assets.

Say: the old design put the risk premium on the *scenario* (8% shock rate). That punished green winners in the shock too — one cause of wrong-direction results. Moving the premium onto the *technology* is what the carbon-premium literature actually measures.

Two modes, because IAMs disagree about prices

Carbon cost = production volume × carbon price × emission factor × (1 – passthrough). The question is *which* emission factor — and that depends on the scenario.

FULL_EF — DEFAULT

Every tonne pays. Used when carbon costs are **not** already inside the scenario's electricity prices.

DIFFERENTIAL_EF

Only emissions above the marginal generator pay. Used when prices already embed carbon — avoids double-counting.

Auto-detection picks the mode from the scenario's carbon-price coverage. Passthrough default 0 — the firm absorbs the full cost (conservative stress-test stance).

Say: the double-counting risk was raised in review as "we need a decision tree" — this *is* the decision tree, shipped and guarded by a runtime warning when modes are mismatched. Note: cost scales with volume (MWh), not capacity.

IAM electricity prices don't clear markets

AR6 prices are LCOE-derived and technology-differentiated — every technology sees its own price. Real markets clear at one price set by the marginal generator.

- Consequence: perpetually negative NPVs for some technologies *in the baseline* — before any shock is applied.
- A model whose baseline says "this plant never made money" cannot measure a transition shock against it.

Say: this is the deepest data problem we have with AR6, and it motivated the MCPR layer on the next slide. It is a property of the scenarios, not of our pipeline.

MCPR v2 — a clearing-price adjustment layer

CARBON_EXPLICIT

For scenarios with explicit carbon prices: gas-anchored clearing price, technology value factors, paired with full_ef carbon costing.

MERIT_ORDER_DECLINE

Clearing price declines with renewable penetration (IMF elasticity). Decline applied to the clearing price *before* value factors — a price floor can't silently negate it.

Status — **two-track**: internal working configuration, validated on 5 / 5 IAM providers.

Externally: an exploratory adjustment pending a methods paper and engineering review.

Say: review consensus was "intuitive but heroic assumptions — explore as a paper." We agree, and we frame it that way externally. Providers validated: COFFEE, GCAM, IMAGE, MESSAGEix-GLOBIOM, WITCH. If AR7/NGFS publish usable clearing prices (expected H2 2026), MCPR becomes validation, not dependency.

Retirement by age — a proxy, stated honestly

Companies retire their oldest assets first in the model. Age proxies carbon intensity and retirement priority. It does **not** claim economic optimization.

- Older plants are typically less efficient and dirtier — the intensity proxy holds.
- But a young inefficient plant can be less economic than an old efficient one — so we don't claim the economics.
- Stranding still happens where it should: the valuation layer zeroes the terminal value of persistently loss-making assets, whatever their age.

Say: upgrade path is plant-level intensity from Climate Trace or AR6 efficiency-by-vintage data (future work). Until then, external claims stay at "age as proxy" — this caveat came out of the deep-dive review and we adopt it.

Direction restored in the large majority — not all

PIPELINE

5 / 5

IAM providers run end-to-end, validation
OK

CARBONTECH

78.5%

of cases show the expected negative
shock

GREENTECH

85.6%

of cases show the expected positive
direction

- Residual wrong-direction cases remain — we do not claim "every scenario."
- Gas is the known borderline: merit-order winner in some pathways, carbon loser in others.

Say: figures are from the post-fix April batch on 5 providers × 3 configurations. A canonical direction-expectation matrix rebuilt from those results is the next deliverable and will replace these numbers if they move.

Future improvements

#	IMPROVEMENT	UNLOCKS
1	Provider-level CapEx audit — which IAM O&M series bundle capital?	Safe re-enabling of new-build CapEx per provider
2	Technology-specific decommissioning ratios (coal ≠ gas ≠ wind)	Defensible retirement cost magnitudes
3	Retirement proxy upgrade: Climate Trace / AR6 plant-level intensity	Replaces age-as-proxy caveat with data
4	AR7 / NGFS reassessment (expected H2 2026)	Possible retirement of MCPR as a dependency
5	Canonical direction-expectation matrix as a regression gate	Every model change tested against expected signs
6	Credit-risk layer: Merton PD on top of NPV	Restores TRISK's PD output at asset-level quality
7	MCPR methods paper with engineering input	Peaker vs baseload dynamics; external credibility

Say: items 1–3 are decided and queued; 4 is a trigger we monitor; 5 is the immediate next deliverable; 6–7 are the research track.

SUMMARY

One sentence per change

- Cash flows, not accounting profits — because markets price cash.
- CapEx made explicit; new-build off until double-counting is ruled out.
- Retirement costs money — and fixing that sign fixed the most directions.
- Risk premium moved from scenario to technology — where the evidence is.
- Declining assets stop growing forever: 0% fossil growth, stranding-aware terminal value.
- Carbon costs auto-detect double-counting; prices get a clearing-price layer, honestly labeled exploratory.

Model 7b7546a · plan of record: docs/superpowers/plans/2026-07-06-altr-reconciliation-and-effectiveness-plan.md